

Day 4 – Model Tuning, Regularization & Reproducible Pipelines

Task 1: Build Fully Reproducible Pipelines

- Built complete Scikit-learn pipelines combining feature engineering, preprocessing, and model training.
- Applied imputation, scaling, and one-hot encoding.
- Used `random_state=42` to make results reproducible.
- Saved the trained Logistic Regression and Random Forest pipelines as `.joblib` files.
- Documented the required library versions and reproduction steps.

Task 2: Hyperparameter Search

- Used `RandomizedSearchCV` with 5-fold `StratifiedKFold` cross-validation.
- Optimized models using the F1-score.
- Tuned Logistic Regression parameters such as `C` and regularization settings.
- Tuned Random Forest parameters including:
 - `n_estimators`
 - `max_depth`
 - `min_samples_leaf`
 - `max_features`
- Used `n_jobs=-1` for faster parallel processing.
- Saved the best-performing model pipelines.

Task 3: Diagnose Overfitting and Underfitting

- Generated learning curves for the selected models.
- Compared training and validation performance to identify overfitting and underfitting.
- Studied the effect of Logistic Regression `C` values on model complexity.
- Identified regularization as a method to reduce overfitting.

- Considered tree controls such as `max_depth` and `min_samples_leaf` to improve Random Forest generalization.

Task 4: Probability Calibration & Threshold Selection

- Applied probability calibration using `CalibratedClassifierCV`.
- Evaluated probability quality using the Brier Score.
- Generated calibration-related evaluation results.
- Tested different classification thresholds instead of relying only on 0.50.
- Selected a threshold based on F1-score performance.
- Compared default and tuned thresholds using confusion matrices and classification metrics.

Task 5: Final Evaluation & Save Artifact

- Evaluated the final calibrated model on the untouched hold-out test set.
- Calculated:
 - Accuracy
 - Precision
 - Recall
 - F1-score
 - ROC-AUC
 - Brier Score
- Generated the final confusion matrix and classification report.
- Saved the complete calibrated model, threshold, metrics, and random state as a `.joblib` artifact.
- Reloaded the saved artifact to verify that it can be used for future inference.
- Tested the saved model on sample data to confirm successful prediction.